CHEM 427/527 · SESSION 6 · TUESDAY, OCTOBER 20, 2026

Batch file handling and automated reporting

Matt Bush, with teaching assistants Lucas Narisawa and Chris Weir

Drafted with assistance by Claude Code. Reviewed and revised by Matt.

LEARNING OBJECTIVES

By the end of today you can

1. Build a file inventory with pathlib, one row per file with the day, setting, and role from its folder and file name, and check its count against the laboratory's own list before any file is read.
2. Write one function that reads a spectrum and returns the value at the analytical wavelength, and apply it to every file in a loop that collects one table.
3. Log each file the loop cannot read, with its name and error, and show that the rows read plus the files logged equal the inventory.
4. Decide whether a file that needed a different reader goes into the result. Write the summary table and the report from the same run.

FIRST, EVERYONE PULLS

Get session 6 with git pull

Bring in session 6

`git pull`

A new folder appears:

`S06-batch-reporting`

- **data/flarelab/**: two day folders and a table
- **data/runs/**: 36 run folders

Session 5 is due today

- Upload it by 1:30 pm.
- Restart and Run All before you upload.

Nothing to download today: every run's files are in the repository.

FINDING EVERY FILE IN A FOLDER

Where does a laboratory keep a sample's facts?

One file in Ferrocyanide Standard 240409/

`LE_240409_d_50_mm_ferrocyanide_counts_v5_FLMS018081_…`

day 240409 · distance setting 50 mm · role ferrocyanide · standard v5

The folder

The day. **39 files** in this one.

The file name

Day, distance setting, role, and version, split on "_".

The laboratory's list

213 rows over four days, **78** for these two. The only place the concentration is written.

WHY COUNT FIRST

How many files does the loop read?

`DATA_DIR.glob("*.txt")`

0

The spectra are one level down. No error.

`DATA_DIR.rglob("*.txt")`

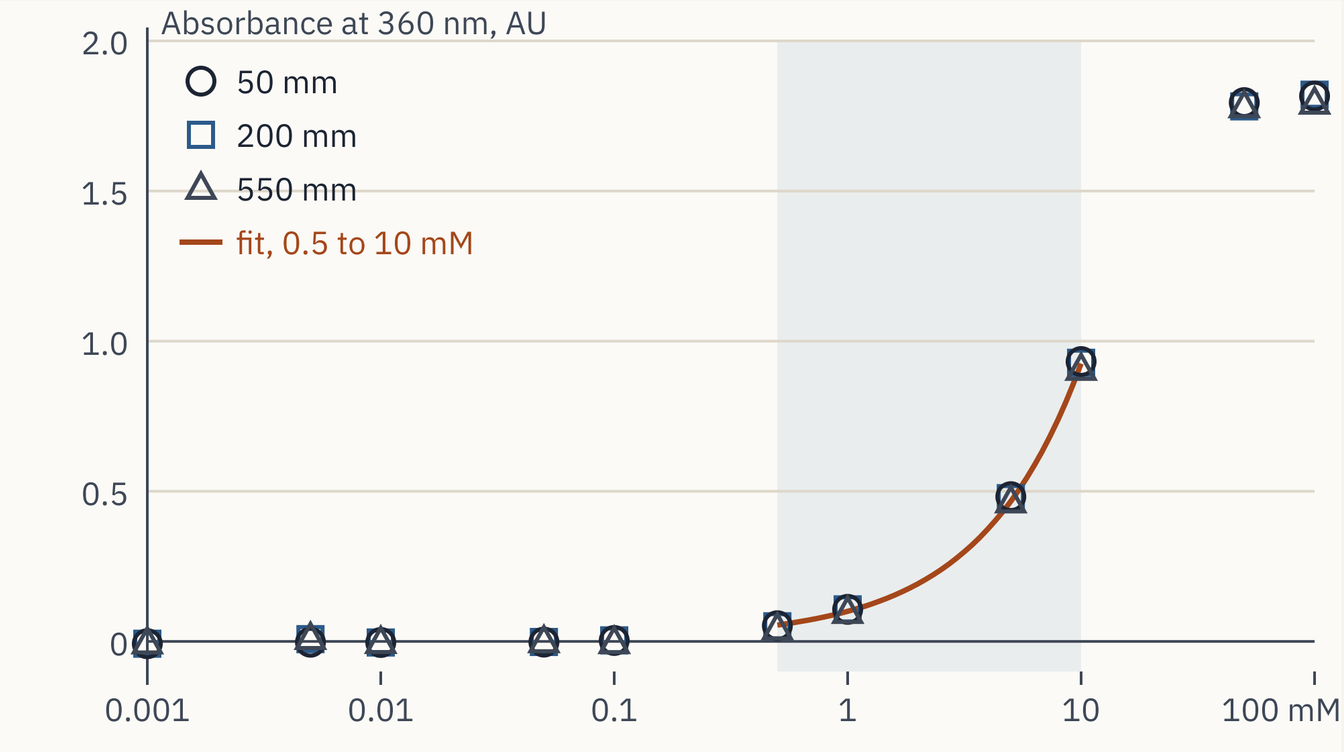
78

Every folder below. The parameter table lists 78 for these two days.

A pattern that matches nothing raises no error. It returns an empty list, and every later cell runs on nothing. Count against the table, 78 against 78, before a spectrum is read.

SECTION 1, THE WORKED EXAMPLE

Three settings, one line



Day 240409, 33 standards. Fit on the 12 from 0.5 to 10 mM: slope **0.0917 AU per mM**, R^2 0.9993.

Day 240410's standards, read against it:

Made up, mM	Recovered
0.5	161.1%
1	94.5%
5	112.8%
10	109.0%

Only 0.5 mM is outside the 85 to 115% window, at every setting: the solution, not one file.

LIVE, ON DEMO/: THREE FILES, ONE WITH FOUR LINES OF TEXT ON TOP

Logging the files the loop cannot read

No try

```
for path in paths:
    read and print
```

```
std_1p0.csv 0.199
std_2p0.csv 0.398
ParserError
```

Stops at the third file. Nothing after it runs.

except: print

```
except Exception as err:
    print(path.name, err)
```

```
std_1p0.csv 0.199
std_2p0.csv 0.398
std_4p0.csv Error tokenizing
data…
```

Two values, one failure named. The loop finishes.

except: continue

```
except Exception as err:
    continue
```

```
std_1p0.csv 0.199
std_2p0.csv 0.398
```

Two values, and no sign of the third file.

Which of these runs would you hand to a grader?

SECTION 2 IS YOURS

Your run: iron in water

The run

- 16 spectra: 7 standards and three waters, three replicates each
- Absorbance at 510 nm, the iron(II) phenanthroline complex
- A sequence table: the file the instrument wrote at each position

Who reads the result

Washington requires a new water system to treat for iron above the **0.3 mg/L** secondary standard (WAC 246-290-320).

Each water in your report is above it or not.

One file in every run raises an error when it is read. A file skipped without a trace is a well never judged. Section 3 asks what to do with it.

ASSIGNMENTS

One run each

#	Run	Waters
1	run_01	spring, river, cistern
2	run_02	spring, cistern, well
3	run_03	tap, river, well
4	run_04	river, cistern, well
5	run_05	spring, cistern, well
6	run_06	tap, river, cistern
7	run_07	spring, river, well
8	run_08	spring, river, well
9	run_09	tap, river, cistern

#	Run	Waters
10	run_10	tap, spring, well
11	run_11	spring, cistern, well
12	run_12	tap, river, well
13	run_13	tap, spring, river
14	run_14	tap, spring, river
15	run_15	spring, river, well
16	run_16	tap, river, cistern
17	run_17	spring, river, cistern
18	run_18	tap, river, cistern

#	Run	Waters
19	run_19	tap, river, cistern
20	run_20	river, cistern, well
21	run_21	tap, river, cistern
22	run_22	tap, spring, river
23	run_23	spring, cistern, well
24	run_24	spring, river, well
25	run_25	tap, spring, river
26	run_26	tap, spring, well

Set **RUN** in Parameters to your run, with the underscore and the leading zero: **run_07**, not run7.

AI PRACTICE 3 OF 8: GIVE THE ASSISTANT A CHECK TO RUN

The printed counts are the check

Live, on demo/, with the try loop selected

What should the except branch do so this loop keeps going when a file will not read? Then **run your version of the loop in my notebook** and **print how many files it read and how many it logged**, next to **the number of files in demo/**.

Before TASK 2, the same prompt on your run, with the number of rows in run_sequence.

The cell it adds prints

2 read

1 logged

3 files in demo/

The counts in the cell's output are evidence. The sentence in the chat that reports them is a claim.

THE JUDGMENT CELL, SECTION 4

Your treatment, and one entry

(a) Your answer

- Which treatment your report uses, and why
- The affected water's concentration, standard deviation, and n under both, and whether either changes its comparison with 0.3 mg/L
- What your summary and report would have said had the except block kept nothing

(b) One judgment entry

- What the assistant suggested
- What you did: changed, rejected, or verified
- Why, from a count or a value in your run

Scored 0 to 2. The prompt before TASK 2 is the source; the counts it printed are the evidence to cite.

SECTION 1 IS READ AND RUN. SECTION 2 IS YOURS.

Go, then submit

Submit

Restart, Run All, save, and upload activity.ipynb by **1:30 pm Thursday, October 22**.

Every run's files are in the repository, so the graders' run reads the same data yours did.

Scoring, 0 to 3

1 if it runs after a restart, **0** if not.

0 to 2 for the judgment cell.

Help

Ask Matt, Lucas, or Chris in the room.

The third hour is an office hour.